

Unsupervised Learning: Dimension Reduction

Why Dimension Reduction?

For Big-Data:

- Data visualization becomes very difficult! (Cannot draw 2D scatterplots between all pairs of features).
- Big-Data often has a high degrees of redundancy. (i.e. correlation among features).
- Many features may be uninformative for the particular problem under study (noise features).
- Dimension reduction ideally allows us retain information on most important features of the data, while reducing noise and simplifying visualization & analysis.

What is Dimension Reduction?

- Map the data into a new low-dimensional space where important characteristics of the data are preserved.
- The new space often gives a (linear or non-linear) transformation of the original data.
- Visualization and analysis (clustering/prediction/...) is then performed in the new space.
- In many cases, (especially for non-linear transformations) interpretation becomes difficult.

Principal Components Analysis (PCA)

PCA

Set-up:

- Data matrix: $\mathbf{X}_{n \times p}$, n observations and p features.

Idea:

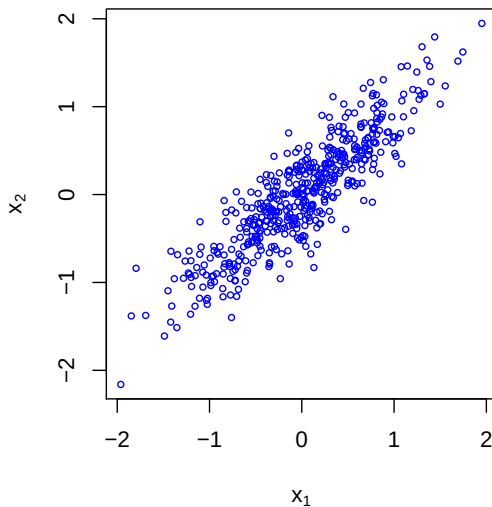
- Not all p features are needed (much redundant info).
- Find low-dimensional linear representations that capture most of the variation in the data.

Uses:

- Ubiquitously used - Dimension reduction, data visualization, pattern recognition, exploratory analysis, etc.
- “Best” linear dimension reduction possible.

PCA - Main Idea

Question: What is a good 1D representation of the data?



PCA - Main Idea

Some Possibilities:

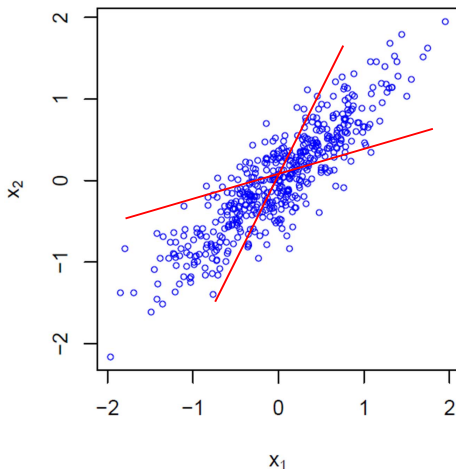
- Use one of the variables (e.g. x_1).
- Better idea: use a linear combination of the variables (i.e. a weighted average).

$$z_1 = v_1 x_1 + v_2 x_2 = \mathbf{v}^T \mathbf{X}$$

How to choose the weights (v_1 and v_2)?

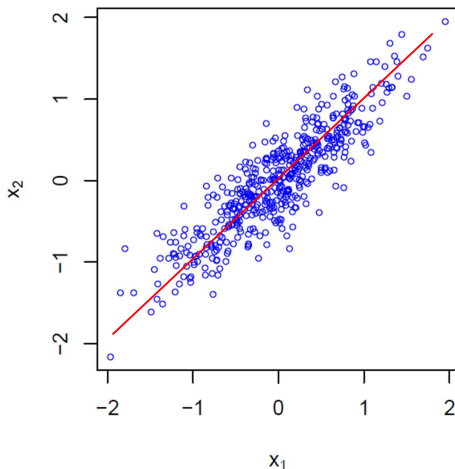
PCA - Main Idea

Many possibilities, but which one is a good choice?



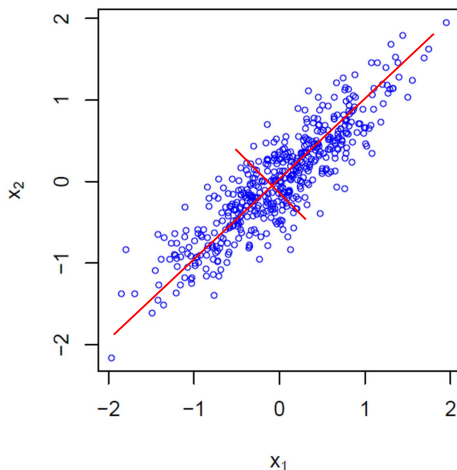
PCA - Main Idea

Find line that maximizes the variance of the data projected onto the line:

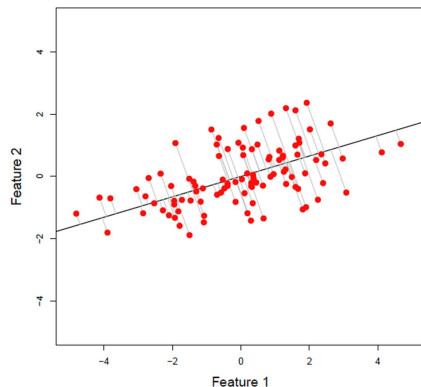


PCA - Main Idea

Subsequent components orthogonal (perpendicular).



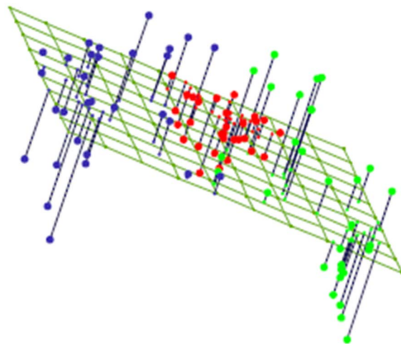
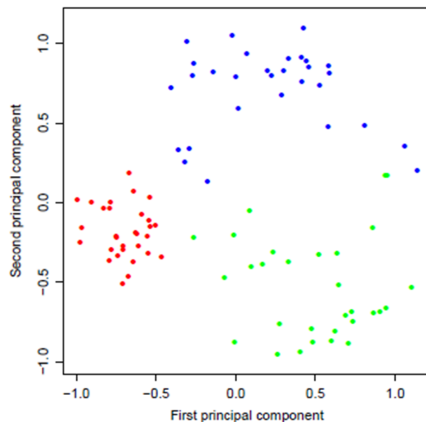
PCA - Main Idea



- PCA minimizes orthogonal projection onto line: $Z = v_1x_1 + v_2x_2$.
- Note: Not same as OLS which minimizes projection of y onto x !

PCA - Main Idea

3D Projection onto a Hyperplane:



PCA - Criterion

PCA Criterion - PC 1 (Population):

$$\underset{\mathbf{v}}{\text{maximize}} \quad \text{Var}(\mathbf{X}\mathbf{v}) \quad \text{subject to } \|\mathbf{v}\|_2 = 1$$

$$\underset{\mathbf{v}}{\text{maximize}} \quad \mathbf{v}^T \text{Var}(\mathbf{X}) \mathbf{v} \quad \text{subject to } \|\mathbf{v}\|_2 = 1$$

$$\underset{\mathbf{v}}{\text{maximize}} \quad \mathbf{v}^T \mathbf{\Sigma} \mathbf{v} \quad \text{subject to } \|\mathbf{v}\|_2 = 1$$

where $\mathbf{\Sigma} = \text{Cov}(\mathbf{X})$.

- Finds linear combination of features that maximizes the variance.

PCA - Criterion

PCA Criterion - PC k (Population):

$$\underset{\mathbf{v}_k}{\text{maximize}} \quad \mathbf{v}_k^T \mathbf{\Sigma} \mathbf{v}_k \quad \text{subject to} \quad \|\mathbf{v}_k\|_2 = 1 \text{ \& } \mathbf{v}_k^T \mathbf{v}_j = 0 \quad \forall j < k.$$

- Subsequent linear combinations are orthogonal to previous combinations.
- **Uncorrelated.**

PCA - Criterion

PCA Criterion - Sample Version:

$$\underset{\mathbf{v}_1, \dots, \mathbf{v}_K}{\text{maximize}} \quad \mathbf{v}_k^T \mathbf{X}^T \mathbf{X} \mathbf{v}_k \quad \text{subject to } \|\mathbf{v}_k\|_2 = 1 \text{ \& } \mathbf{v}_k^T \mathbf{v}_j = 0 \quad \forall j < k.$$

Replaces Σ with estimate $\mathbf{X}^T \mathbf{X} / n$.

Solution: Eigenvalue decomposition of $\mathbf{X}^T \mathbf{X}$. (`eigen()` in R)

PCA - Criterion

Equivalent PCA Criterion:

$$\begin{aligned} & \underset{\mathbf{u}_1, \dots, \mathbf{u}_K, \mathbf{v}_1, \dots, \mathbf{v}_K}{\text{maximize}} \quad \mathbf{u}_k^T \mathbf{X} \mathbf{v}_k \quad \text{subject to } \|\mathbf{v}_k\|_2 = 1 \ \& \ \mathbf{v}_k^T \mathbf{v}_j = 0 \ \forall j < k. \\ & \|\mathbf{u}_k\|_2 = 1 \ \& \ \mathbf{u}_k^T \mathbf{u}_j = 0 \ \forall j < k. \end{aligned}$$

- Finds left and right projection that maximize variance.

Solution: Singular Value Decomposition (SVD) of $\mathbf{X}$. (`svd()` in R)

PCA - Parts of the Solution

$$\text{SVD: } \mathbf{X}_{n \times p} = \mathbf{U}_{n \times n} \mathbf{D}_{n \times p} \mathbf{V}_{p \times p}^T$$

- Singular vectors: (left) $\mathbf{U}$ and (right) $\mathbf{V}$.
 - ▶ Orthonormal - $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ and $\mathbf{V}^T \mathbf{V} = \mathbf{I}$.
- Singular values: Diagonals of $\mathbf{D}$.
 - ▶ $d_1 \geq d_2 \geq \dots \geq d_r$ where $r = \text{rank}(\mathbf{X})$.

SVD Solution to PCA:

- **PCs:** $\mathbf{Z} = \mathbf{X}\mathbf{V}$ or $\mathbf{Z} = \mathbf{U}\mathbf{D}$. ($\mathbf{U}$ are un-scaled PCs).
 - ▶ $\mathbf{z}_k = \mathbf{X}\mathbf{v}_k$ - k^{th} PC.
 - ▶ $\mathbf{z}_1 \dots \mathbf{z}_K$ gives best K -dimensional projection of the data.
- **PC Loadings:** $\mathbf{V}$.
 - ▶ $\mathbf{v}_k$ - k^{th} PC loading (feature weights).

PCA - Properties

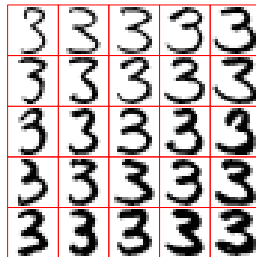
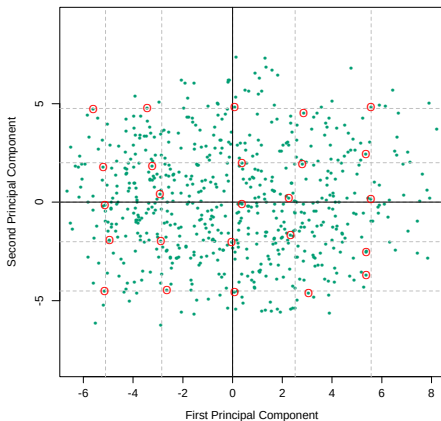
- Unique.
 - ▶ $\mathbf{U}$ and $\mathbf{V}$ unique up to a sign change.
 - ▶ $\mathbf{D}$ unique.
- Nested, Ordered components.
 - ▶ $d_1 \geq d_2 \geq \dots d_p$.
- Orthogonal.
 - ▶ $\mathbf{U}$ and $\mathbf{V}$ orthogonal.
- Global Solution.

PCA - Pattern Recognition

- $\mathbf{u}_1$ - first column of $\mathbf{U}$ encodes first major pattern in observation space.
- $\mathbf{v}_1$ - first column of $\mathbf{V}$ encodes the associated first pattern in feature space.
- d_1 gives strength of first pattern.
- Subsequent patterns are **uncorrelated** to first pattern (i.e. orthogonal).
- $\mathbf{X} \approx \sum_{k=1}^K d_k \mathbf{u}_k \mathbf{v}_k^T$ - data is comprised of a series of patterns.

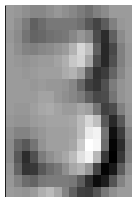
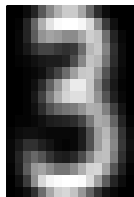
PCA - Pattern Recognition

Patterns in observation space:



PCA - Pattern Recognition

Patterns in feature space:



Discussion

- What pattern does the first PC find?
- What pattern does the second PC find?
- Questions?

PCA - Data Visualization

PC Scatterplots:

- Problem: Can't visualize all the features.
- Solution: Plot $\mathbf{u}_1$ vs. $\mathbf{u}_2$ and so forth.
- Advantages:
 - ▶ Dramatically reduces number of 2D scatterplots to visualize.
 - ▶ Focuses on patterns with most variance.

PC Loadings Plots:

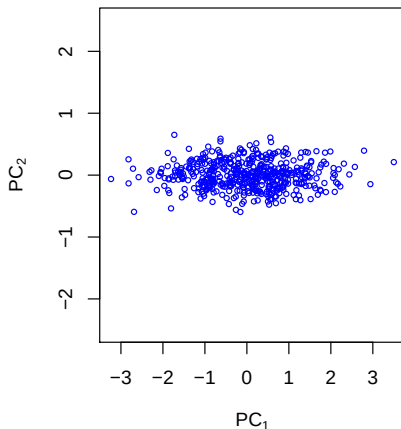
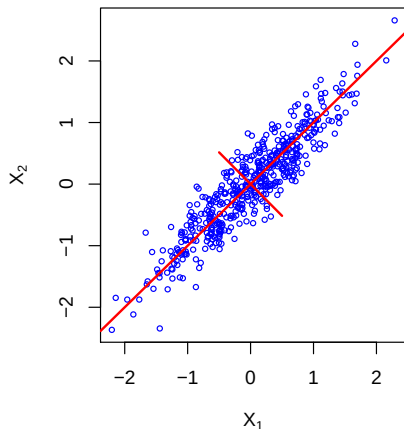
- Scatterplots of $\mathbf{v}_1$ vs. $\mathbf{v}_2$.
- Visualizations of $\mathbf{v}_k$.

Biplot:

- Scatterplot of PC 1 vs. PC 2 with loadings of $\mathbf{v}_1$ vs. $\mathbf{v}_2$ overlaid.

PCA - Data Visualization

Scatterplots:



- Plotting Scatterplot PCs roughly equivalent to rotating axes of original plot.

PCA - Dimension Reduction

Best low-rank approximation to the data:

$$\underset{\tilde{\mathbf{X}}}{\text{minimize}} \quad \|\mathbf{X} - \tilde{\mathbf{X}}\|_F^2 \quad \text{subject to } \text{rank}(\tilde{\mathbf{X}}) = K$$

Solution: $\tilde{\mathbf{X}} = \sum_{k=1}^K d_k \mathbf{u}_k \mathbf{v}_k^T$ - SVD / PCA solution!

- PCA also finds best data compression to minimize reconstruction error.
- PCA yields “best” linear dimension reduction possible!

PCA - Dimension Reduction

How much variance is explained? (i.e. extent of dimension reduction)

- Variance explained by k^{th} PC:

$$d_k^2 = \mathbf{v}_k^T \mathbf{X}^T \mathbf{X} \mathbf{v}_k.$$

- Total variance of data:

$$\sum_{j=1}^p d_j^2.$$

- Proportion of variance explained by k^{th} PC:

$$d_k^2 / \sum_{j=1}^p d_j^2.$$

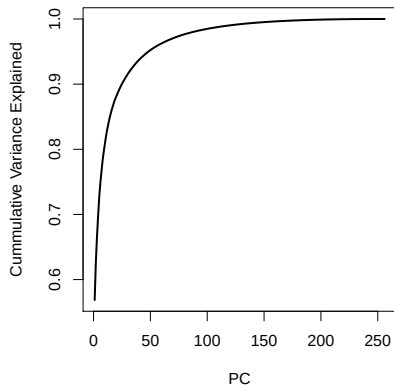
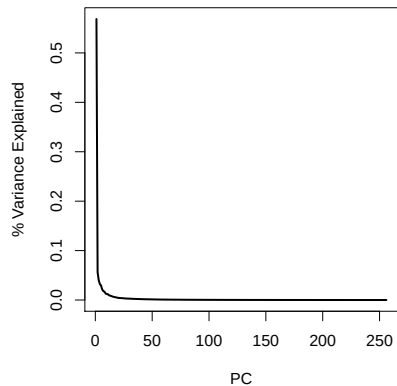
- Cumulative variance explained by first r PCs:

$$\sum_{k=1}^r d_k^2 / \sum_{j=1}^p d_j^2.$$

(Extent of dimension reduction achieved by first r PC projections.)

PCA - Dimension Reduction

Screeplot:



PCA - Dimension Reduction

How to choose K ?

- Elbow in screeplot.
- Take K that explains at least 90% (95%, 99%, etc.) variance.
- Parallel analysis: keep PCs whose eigenvalues exceed those from a “null” dataset with no low-dimensional structure.
- More sophisticated:
 - ▶ Cross-Validation done internally.
 - ▶ Validation via matrix completion.
 - ▶ Nuclear norm penalties.

PCA - Center and Scale?

- Typically, one should center features (i.e. columns of $\mathbf{X}$).
 - ▶ Maximizing variance interpretation (assumes multivariate Gaussian model).
- Scaling changes PCA solution.
 - ▶ Features with large scale contribute more to variance, have large PC loadings.

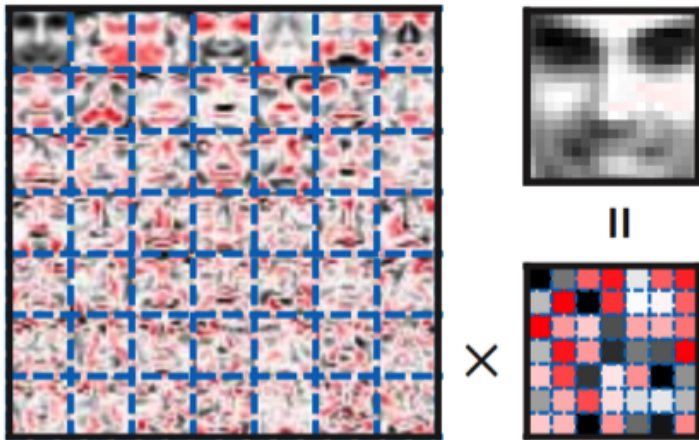
General Suggestions:

- Scale if features measured differently. (Example - US college data).
- Don't scale if features measured in same way & scale has meaning. (Example - gene expression data).

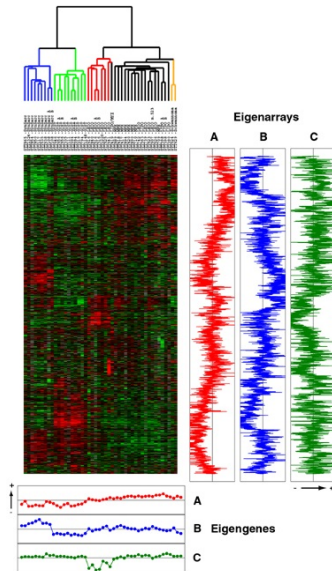
PCA - Applications

“EigenImages” or “EigenFaces”

PCA



PCA - Applications



PCA - Summary

Strengths:

- “Best” linear dimension reduction.
- Interpretation:
 - ▶ Ordered / orthogonal components.
 - ▶ Unique, global solution.
- Data visualization.
- Pattern recognition.
- Others?

Weaknesses:

- Non-linear patterns.
- Others?

PCA - Best Practices

When someone gives you a data matrix ...

Apply PCA first!

Discussion

Quick Quiz (T/F):

- You should always center and scale before applying PCA.
- PCA will perform poorly if your data doesn't lie on a line.
- You can only visualize the first two PCs.
- PCA only finds patterns amongst the observations.
- PC i always contains a more important pattern than PC $i + 1$.

Discussion:

- How will you use PCA for your work?
- Questions?

PCA Extensions

Sparse PCA

Motivation:

- When $p \gg n$, many features irrelevant.
- PCA can perform poorly.

Idea:

- Sparsity in $\mathbf{V}$: zero out irrelevant features from PC loadings.
- Advantage: Find important features that contribute to major patterns in the data.

How?

- Typically, optimize PCA criterion with sparsity-encouraging penalty of $\mathbf{V}$.
- Many methods - active area of research!

In R: SPC in PMA package.

Functional PCA

Motivation:

- Times series, ordered data, spatial data.

Idea:

- Want PC loadings to be smooth (vary continuously) over time or space.
- Advantage: Improve interpretation.

How?

- Typically, optimize PCA criterion with a penalty that encourages smoothness of $\mathbf{V}$ over time or space.
- Many methods for both functional data (data in the form of curves) and discretely-sampled functional data (e.g. discrete time points or specific locations).

In R: package `fpca`.

Kernel PCA

Motivation:

- Non-linear patterns.

Idea:

- Embed inner product distances $(x_i^T x_{i'})$ in a higher-dimensional “kernel” space, $k(x_i, x_{i'})$.
- Kernel examples:
 - ▶ Radial: $k(x_i, x_{i'}) = e^{-\|x_i - x_{i'}\|_2^2 / 2\sigma^2}$.
 - ▶ Polynomial: $k(x_i, x_{i'}) = (cx_i^T x_{i'} + 1)^d$.
- Kernel Matrix: $\mathbf{K}_{n \times n} : \mathbf{K}_{ii'} = k(x_i, x_{i'})$.
 - ▶ Idea: $\mathbf{K}$ a non-linear distance matrix.
- Find major non-linear patterns by performing PCA on $\mathbf{K}$:

$$\mathbf{K} = \mathbf{U} \mathbf{D}^2 \mathbf{U}^T$$

Supervised Dimension Reduction

Partial Least Squares:

- Best dimension reduction of cross-covariance between $\mathbf{X}$ and $\mathbf{Y}$ such that factors are orthogonal to $\mathbf{X}$.

Canonical Correlations Analysis:

- Best dimension reduction of cross-covariance between $\mathbf{X}$ and $\mathbf{Y}$ such that bi-projection is orthogonal to $\mathbf{X}$ or $\mathbf{Y}$.

Linear Discriminant Analysis (classification):

- Best dimension reduction of between class covariance matrix relative to within class covariance.

Non-Negative Matrix Factorization (NMF)

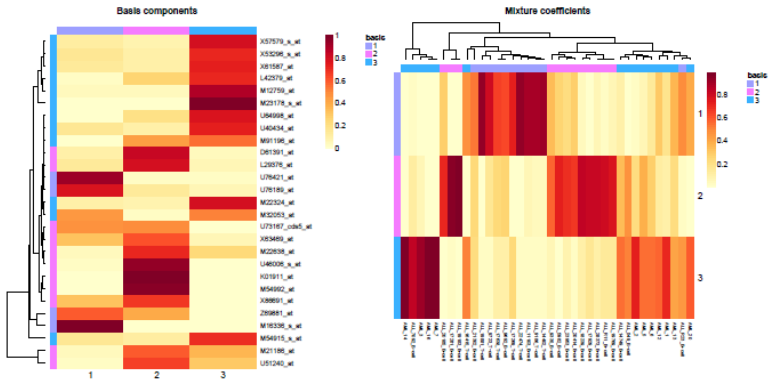
NMF

Idea: $\mathbf{X}_{n \times p} \approx \mathbf{W}_{n \times K} \mathbf{H}_{K \times p} = \sum_{k=1}^K \mathbf{W}_{:,k} \mathbf{H}_{k,:}$ with $K \ll p$.

- $\mathbf{X} \geq 0$ - non-negative data matrix.
- $\mathbf{W} \geq 0$ - non-negative observation factors; often sparse (Basis Factors).
 - ▶ $\mathbf{W}_{:,k} \geq 0$ - k^{th} observation factor.
- $\mathbf{H}_{kj} \geq 0$ - non-negative feature factors; often sparse (Mixture Factors).
 - ▶ $\mathbf{H}_{k,:} \geq 0$ - mixture of features that comprise the k^{th} factor.

Like PCA except finds patterns with same direction of correlation.

NMF



NMF Interpretation

Topic Modeling:

- $\mathbf{X}$ a matrix of news articles (rows) by words (columns) whose entries are word counts.
 - ▶ $\mathbf{X} \approx \sum_{k=1}^K \mathbf{W}_{:,k} \mathbf{H}_{k,:}$ - sum of topics.
 - ▶ $\mathbf{X}_{ij} = \mathbf{W}_{i,:}^T \mathbf{H}_{:,j} = \sum_{k=1}^K \mathbf{W}_{ik} \mathbf{H}_{kj}$.
- Topic k : Outer-product of k^{th} column of $\mathbf{W}$ ($\mathbf{W}_{:,k}$) and k^{th} row of $\mathbf{H}$ ($\mathbf{H}_{k,:}$).
 - ▶ E.g. Oil prices.
- $\mathbf{H}_{k,:}$ non-zeros- words contributing to topic k .
 - ▶ E.g. Strait of Hormuz, gas, crude, Iran etc.
- $\mathbf{W}_{:,k}$ non-zeros - news articles belonging to topic k .
 - ▶ E.g. "Global oil prices jump as war threats intensify" (*New York Times*).

NMF Criterion - Continuous Data

$$\begin{aligned} & \underset{\mathbf{W}, \mathbf{H}}{\text{minimize}} \quad \|\mathbf{X} - \mathbf{W} \mathbf{H}\|_F^2 \\ & \text{subject to} \quad \mathbf{W}_{ik} \geq 0 \ \& \ \mathbf{H}_{kj} \geq 0 \end{aligned}$$

(PCA criterion except with non-negativity constraints.)

Algorithm Updates: (Alternating Non-negative Least Squares)

$$\begin{aligned} \hat{\mathbf{W}} &= \left(\mathbf{X} \mathbf{H}^T (\mathbf{H} \mathbf{H}^T)^{-1} \right)_+ \\ \hat{\mathbf{H}} &= \left((\mathbf{W}^T \mathbf{W})^{-1} \mathbf{W}^T \mathbf{X} \right)_+ \end{aligned}$$

Local Solution.

NMF Criterion - Count Data

$$\begin{aligned} & \underset{\mathbf{W}, \mathbf{H}}{\text{minimize}} && \sum_{i=1}^n \sum_{j=1}^p [\mathbf{x}_{ij} \log(\mathbf{W} \mathbf{H})_{ij} - (\mathbf{W} \mathbf{H})_{ij}] \\ & \text{subject to} && \mathbf{W}_{ik} \geq 0 \ \& \ \mathbf{H}_{kj} \geq 0 \end{aligned}$$

Algorithm Updates:

$$\begin{aligned} \hat{\mathbf{W}}_{ik} &= \hat{\mathbf{W}}_{ik} \left(\frac{\sum_{j=1}^p \hat{\mathbf{H}}_{kj} \mathbf{x}_{ij} / (\hat{\mathbf{W}}^T \hat{\mathbf{H}})_{ij}}{\sum_{j=1}^p \hat{\mathbf{H}}_{kj}} \right) \\ \hat{\mathbf{H}}_{kj} &= \hat{\mathbf{H}}_{kj} \left(\frac{\sum_{i=1}^n \hat{\mathbf{W}}_{ik} \mathbf{x}_{ij} / (\hat{\mathbf{W}}^T \hat{\mathbf{H}})_{ij}}{\sum_{i=1}^n \hat{\mathbf{W}}_{ik}} \right) \end{aligned}$$

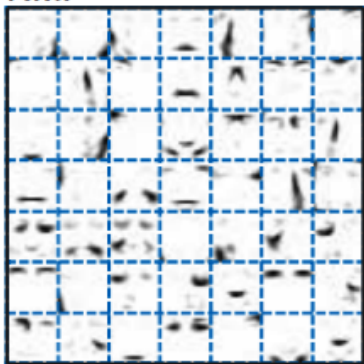
Local solution.

NMF - Uses

- ① Dimension Reduction / Pattern Recognition.
 - ▶ Similar to PCA (e.g. component scatterplots) except that patterns of correlation found in the same direction.
- ② Archetypal Analysis.
 - ▶ Caricatures (segments; contrastive categorization) vs. Prototypes (averages).
- ③ Soft-clustering.
 - ▶ Discussed Next Lecture!

NMF - Archetypal Analysis

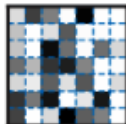
NMF



Original



$\times$

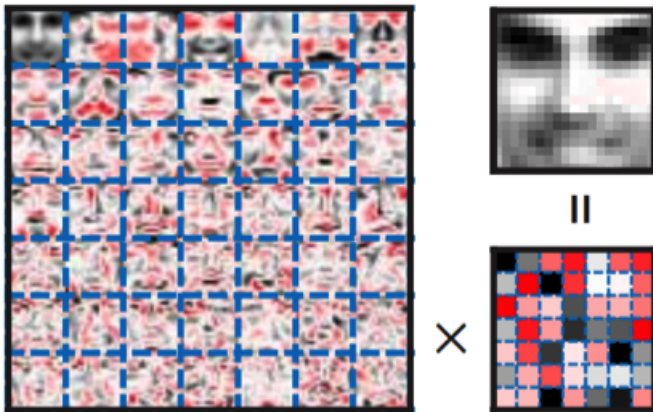


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NMF - Archetypal Analysis

PCA



PCA vs. NMF

Similarities:

- Linear Dimension Reduction.
- Interpretation.

Differences:

- Factors are unordered.
- Factors NOT orthogonal.
- Changing K can fundamentally change factors.
- Non-unique, non-global solution.
- Depends on initialization. (Run several times and take the best).

Choosing K

Choice depends on goal:

- Dimension Reduction:
 - ▶ Residual sums of squares (or dispersion) - Screeplot.
- Clustering:
 - ▶ Consensus, silhouette, etc. (Discussed next lecture!).
- Archetypal Analysis:
 - ▶ Sparsity, factor purity, etc.

NMF - Summary

Strengths:

- Interpretation (often more appealing than PCA!).
- Applications - Clustering & Archetypal Analysis.
- Pattern Recognition.
- Others?

Weaknesses:

- Local solutions that depend strongly on K .
- Others?

In R: NMF package.

Independent Components Analysis (ICA)

Pre-processing Step: Reduce $\mathbf{X}_{n \times p}$ to $\tilde{\mathbf{X}}_{K \times p}$ with $K < n$ # independent sources. (Typically via PCA!)

Idea: $\tilde{\mathbf{X}}_{K \times p} \approx \mathbf{A}_{K \times K} \mathbf{S}_{K \times p}$.

- Assumption: $\tilde{\mathbf{X}}$ a matrix of K scrambled independent signals.
- $\mathbf{A}_{K \times K}$ *Mixing Matrix* - denotes how signals are scrambled to form sources in data.
- $\mathbf{S}_{K \times p}$ *Signal Matrix* - each row of $\mathbf{S}$ is an independent signal.

PCA finds uncorrelated, but not independent signals.

ICA Uses

① Blind Source Separation.

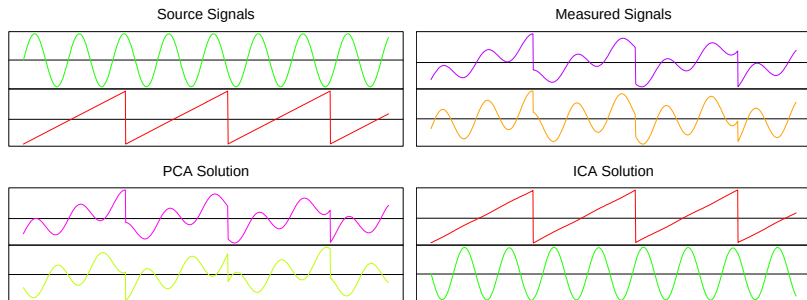
- ▶ Assume K independent signals got scrambled, but record K scrambled versions of the signal.
- ▶ Cocktail Party Problem.

② Denoising.

- ▶ Noise - independent from true signals.

ICA vs. PCA

Blind Source Separation:



ICA Algorithms

Fast ICA:

- Finds projections of $\mathbf{X}$ that are “non-Gaussian”.
- Uses non-Gaussian contrast functions:
 - ▶ $g(x) = x^4$.
 - ▶ $g(x) = \tanh(x)$.
- Generalization of projection pursuit.

Others:

- Infomax (entropy).

Not Statistically Independent!

PCA vs. ICA

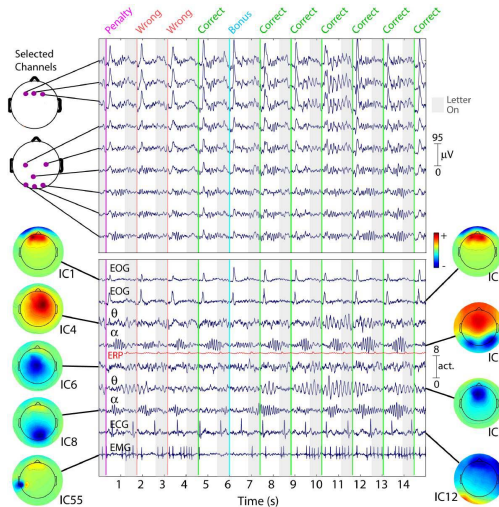
Similarities:

- Linear Dimension Reduction.
- Interpretation.

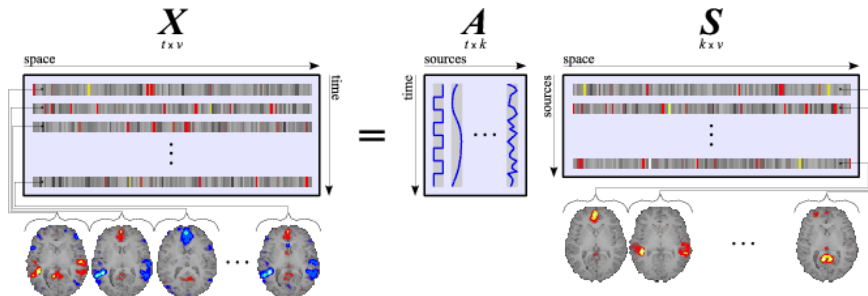
Differences:

- Factors are unordered.
- Factors NOT invariant - same solution by applying a permutation.
- Factors NOT orthogonal.
- Changing K can fundamentally change factors.
- Non-unique.
- No optimization criterion to evaluate solution.

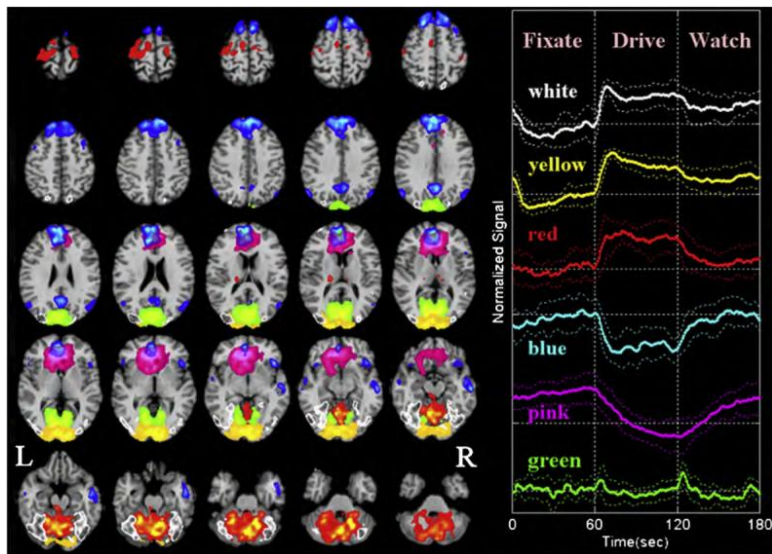
ICA Applications - EEG



ICA Applications - fMRI



ICA Applications - fMRI



ICA Summary

Strengths:

- Interpretation.
- Applications - Blind Source Separation & Denoising.
- Others?

Weaknesses:

- Solutions that depend strongly on K .
- Solutions can be rotated.
- Others?

In R: `fastICA` package.

Multidimensional Scaling (MDS)

Multidimensional Scaling (MDS)

Idea:

- Visually represent proximities (similarities or distances) between objects in a lower dimensional space.
- Input: Matrix of similarities or dissimilarities, $\mathbf{D}_{n \times n}$ (don't need the data itself!).
- Goal: Find projections ($\mathbf{z}_1, \dots, \mathbf{z}_K$ where $\mathbf{z} \in \mathbb{R}^n$) that preserve original distances in $\mathbf{D}$ in a lower dimensional space ($K \ll n$).
- 2 Types: Classical (Metric) MDS and Non-metric MDS.
- Non-linear dimension reduction.

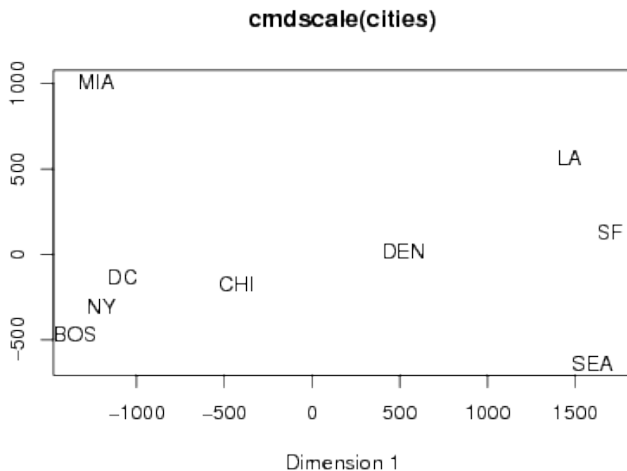
MDS - Example

Consider the distances between nine American cities:

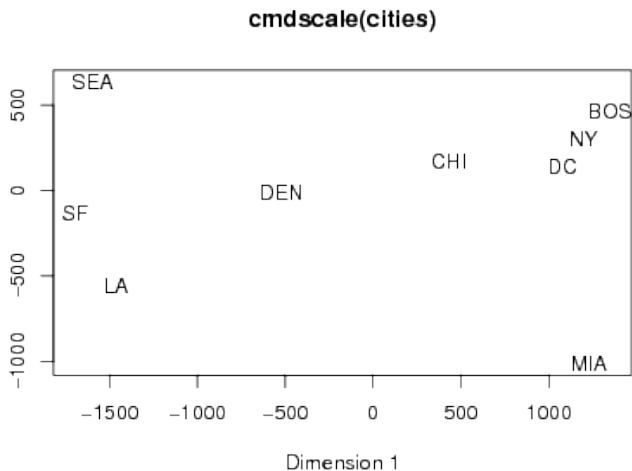
	BOS	CHI	DC	DEN	LA	MIA	NY	SEA	SF
BOS	0	963	429	1949	2979	1504	206	2976	3095
CHI	963	0	671	996	2054	1329	802	2013	2142
DC	429	671	0	1616	2631	1075	233	2684	2799
DEN	1949	996	1616	0	1059	2037	1771	1307	1235
LA	2979	2054	2631	1059	0	2687	2786	1131	379
MIA	1504	1329	1075	2037	2687	0	1308	3273	3053
NY	206	802	233	1771	2786	1308	0	2815	2934
SEA	2976	2013	2684	1307	1131	3273	2815	0	808
SF	3095	2142	2799	1235	379	3053	2934	808	0

Can we represent these cities in a 2D space like a map?

MDS - Example



MDS - Example



MDS - Example



Classical (Metric) MDS

- Idea: Perform PCA (eigenvalue decomposition) on the doubly centered distance matrix, $\mathbf{D}$.
 - ▶ $\mathbf{H} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the centering matrix.
 - ▶ $\mathbf{z}_1, \dots, \mathbf{z}_K$ are the top K PCs of $\mathbf{H}\mathbf{D}\mathbf{H}$.
- Fact: When $\mathbf{D}$ is the matrix of Euclidean distances, classical MDS is equivalent to PCA.
 - ▶ Differences for other distance metrics.

In R: `cmdscale`.

Metric MDS

- Idea: Optimize stress function that keeps distances in $\mathbf{Z}$ close to that of $\mathbf{D}$.

Stress Functions:

- Least squares or Kruskal-Shephard Scaling:

$$S_D(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_K) = \sqrt{\sum_{i \neq i'} (d_{ii'} - \|\mathbf{z}_i - \mathbf{z}_{i'}\|)^2}.$$

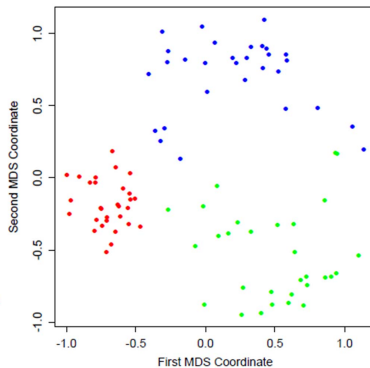
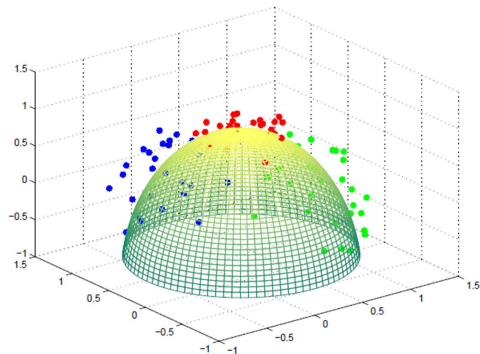
- Sammon mapping: preserve smaller pairwise distances

$$\sum_{i \neq i'} \frac{(d_{ii'} - \|\mathbf{z}_i - \mathbf{z}_{i'}\|)^2}{d_{ii'}}.$$

- Shepard-Kruskal metric scaling ($\theta(\cdot)$: an increasing function):

$$\frac{\sum_{ii'} [\theta(\|\mathbf{z}_i - \mathbf{z}_{i'}\|) - d_{ii'}]^2}{\sum_{ii'} d_{ii'}^2}.$$

MDS - Example



MDS Properties

- Data not needed - only dissimilarities.
- Choosing K :
 - ▶ Scree plot (like PCA).
 - ▶ Shepard Diagram - plot proximities against distances in Z .
- Interpreting MDS maps:
 - ▶ Axes and orientation arbitrary.
 - ▶ Can be rotated.
 - ▶ Only relative locations important.
 - ▶ Typically looks for objects close in the MDS map.
 - ▶ Can be distortions.

MDS vs. PCA

Similarities:

- Dimension reduction for visualization.

Differences:

- Non-linear vs. Linear.
- Local solution & arbitrary map.
- Non-unique & local solution.
- Only yields visualization / patterns among n objects.

MDS - Summary

Strengths:

- Visualizing proximities.
- Only need dissimilarities.
- Non-linear dimension reduction.
- Others?

Weaknesses:

- Arbitrary maps.
- High-dimensional settings? ($p \gg n$ - more features than objects)
- Others?

In R: `dist`; `cmdscale` - classical MDS; `isoMDS` - Kruskals's MDS and `sammon` in MASS package.

Neighbor Embeddings: tSNE & UMAP

Neighbor Embeddings

Idea:

- Find lower dimensional embedding that preserves relationships between “close neighbors”.
- Non-linear dimension reduction (manifold learning).
- Useful for visualizations and clustering.
- Works well for high-dimensional data ($p \gg n$).

Neighbor Embeddings

Core Algorithm:

- ① Represent high-dimensional similarities - $p_{i,i'}$.
 - ▶ tSNE - normalized Gaussian kernel (perplexity hyperparameter controls kernel width).
 - ▶ UMAP - Gaussian kernel (K -nearest neighbor weights; K a hyperparameter)
- ② Represent low-dimensional similarities - $q_{i,i'} = q(z_i, z_{i'})$
 - ▶ tSNE - Cauchy (or t) kernel.
 - ▶ UMAP - Generalized Cauchy kernel.
- ③ Optimize for z 's that minimize discrepancy between p 's and q 's.
 - ▶ tSNE - Kullback-Leibler Divergence.
 - ▶ UMAP - Cross Entropy Loss.

t-Stochastic Neighbor Embedding (tSNE)

- Set up: Represent data, $x_1, \dots, x_n \in \mathbb{R}^p$, in a lower dimensional space, $z_1, \dots, z_n \in \mathbb{R}^k$ where $k \ll p$ (usually $k = 2$).
- Data Similarity (normalized Gaussian kernel; σ related to perplexity):

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2}, \quad p_{j|i} = \frac{\exp(-(x_i - x_j)^2 / 2\sigma^2)}{\sum_{k \neq i} \exp(-(x_i - x_k)^2 / 2\sigma^2)}$$

- Lower-Dimensional Similarity (normalized t/Cauchy kernel):

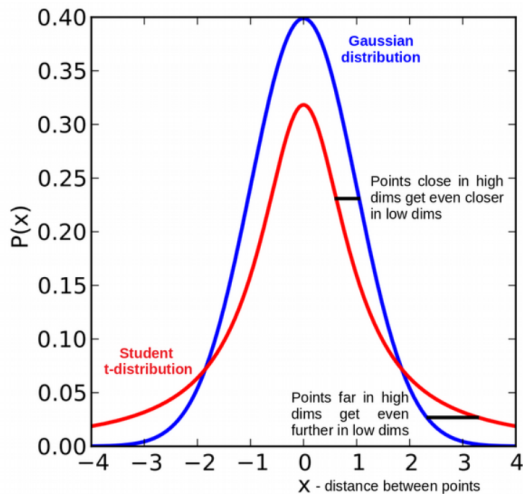
$$q(z_i, z_j) = \frac{g(|z_i - z_j|)}{\sum_{k \neq i} g(|z_i - z_k|)}, \quad g(z) = \frac{1}{1 + z^2}.$$

- Minimize Kullback-Leibler divergence:

$$\underset{z_1, z_2, \dots, z_n}{\text{minimize}} \quad \sum_{i,j} p_{ij} \log \left(\frac{p_{ij}}{q(z_i, z_j)} \right)$$

- Gradient descent (with random initialization).

t-Stochastic Neighbor Embedding (tSNE)



UMAP

Similar to tSNE, but with a few differences:

- Data Similarity (Gaussian kernel, general distances, K-Nearest Neighbors, no normalization):

$$p_{ij} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}, \quad p_{i|j} = \exp(-d(x_i, x_j)/\sigma_i).$$

- Lower-Dimensional Similarity (generalized Cauchy kernel):

$$q(z_i, z_j) = \frac{1}{1 + a(z_i - z_j)^{2b}}, \quad a > 1, b < 1 \text{ (thicker tails)}.$$

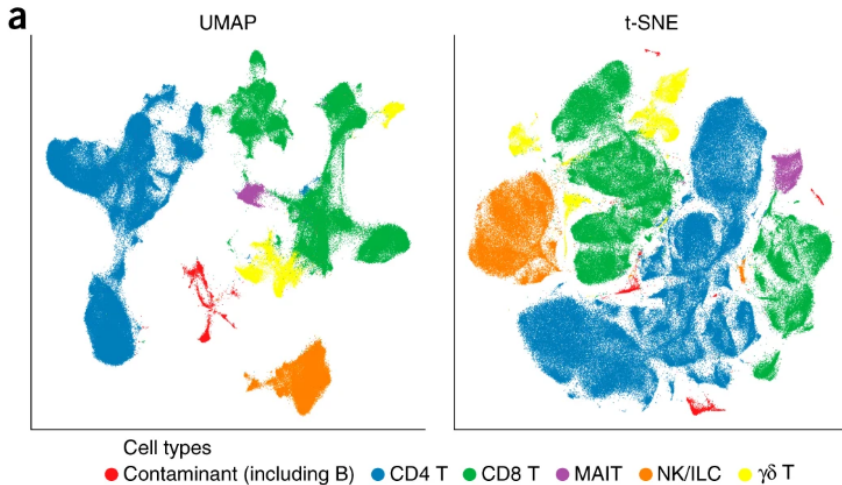
- Minimize cross-entropy:

$$\underset{z_1, z_2, \dots, z_n}{\text{minimize}} \quad \sum_{i,j} p_{ij} \log \left(\frac{p_{ij}}{q(z_i, z_j)} \right) + (1 - p_{ij}) \log \left(\frac{1 - p_{ij}}{1 - q(z_i, z_j)} \right)$$

- Stochastic gradient descent with spectral clustering initialization.

Overall: UMAP better preserves global structures & more tightly packs close neighbors.

tSNE & UMAP - Applications

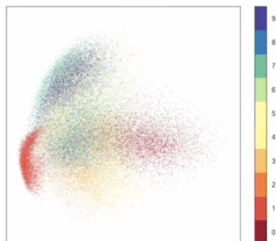


Clustering Cell Types in Single Cell RNA-seq.

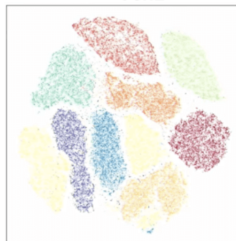
Global vs. Local Interpretation

MNIST Digits

PCA



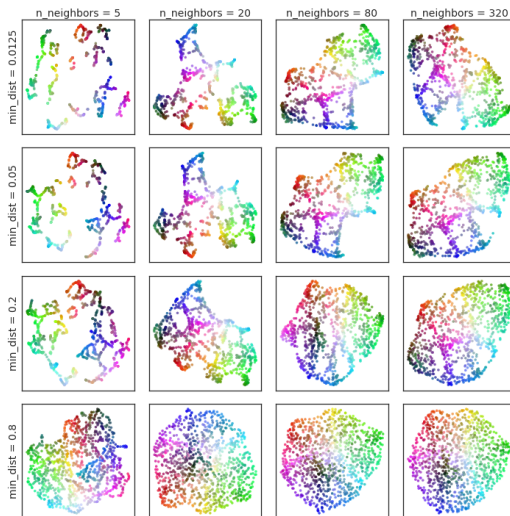
t-SNE



UMAP



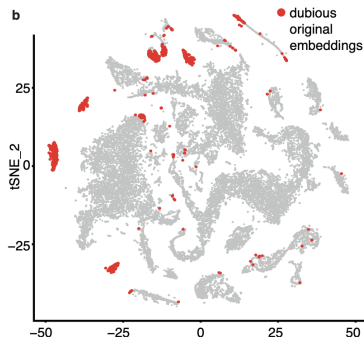
Hyperparameter Tuning



- Very sensitive to hyperparameters!

Hyperparameter Tuning

Stability / Reliability of Neighbors:



- Calculate the reliability / stability of each observation's neighbors from high to low dimensional space.
- Example: scDEED method (Xia et al., 2024)

tSNE & UMAP - Summary

Strengths:

- Data visualization.
- Clustering.
- Non-linear dimension reduction.
- Preserves relationships between close neighbors.

Weaknesses:

- Local, non-unique solutions.
- Patterns only amongst observations.
- Sensitive to hyperparameters.
- Can have distortions.
- Others?

Dimension Reduction & Visualization Wrap-Up

Techniques Covered:

- PCA.
- NMF.
- ICA.
- MDS.
- SNE - tSNE & UMAP.

Discussion

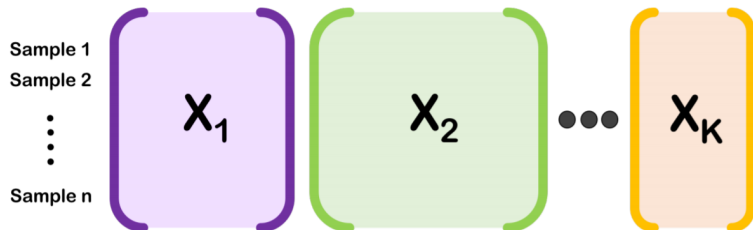
Questions:

- Which techniques are useful for finding patterns amongst observations AND features? Just observations?
- List some advantages of PCA over MDS / tSNE / UMAP. List some disadvantages.
- When would you want use each of the techniques?
- Why do we need so many dimension reduction and visualization techniques?
- If you had new (test) data, how would you use this to validate the patterns you discovered?

Research Highlight

Integrated PCA

Multi-View Data:



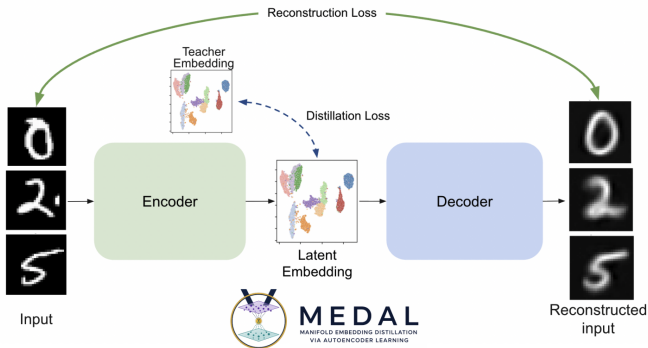
Goal: How do you find joint patterns across the shared subjects?

Solution: iPCA! (Tang and Allen, 2021)

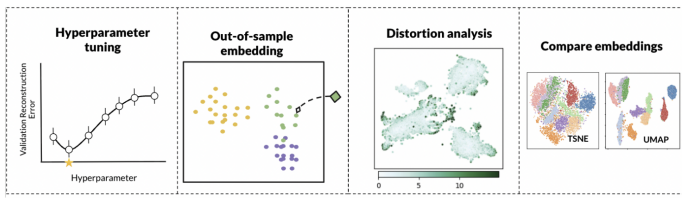
- Generalization of PCA using matrix-variate normal.

Research Highlight

A

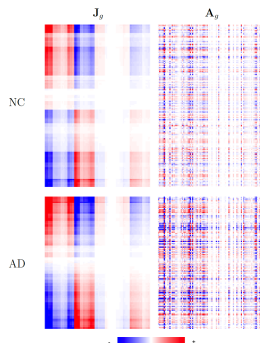


B



Chang, Zikry, and Allen, 2026

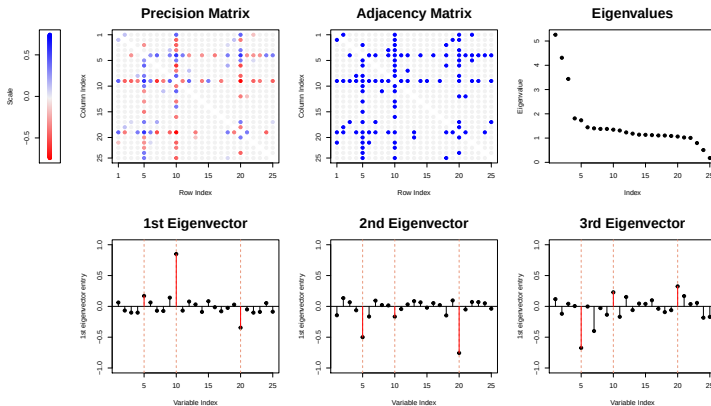
Multi-Group PCA and beyond



Solution: Joint and Individual Component Regression (JICO) (Wang et al., JCGS, 24)

- Generalization of PCA for multi-group data using continuum regression framework; covers JIVE (Lock et al., 13) as a special case.

Hubs and Precision Matrix



Inverse Principal Components for Hub Detection (IPC-HD) (Sanchez Gomez, Mo, Zhao, and Liu, JASA, 2025).